

LLM-Guided Data Provenance

Human-Centric Transparency for Big Data Pipelines

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1. Motivation and Problem

Modern data pipelines generate massive datasets from diverse sources, creating a strong demand for reliable lineage tracking. While solutions like OpenLineage capture lineage as Directed Acyclic Graphs (DAGs), analysts often struggle to interpret them due to data engineering errors, intentional obfuscation, or sheer complexity.

We introduce DAG-to-RAG, a LLM-guided provenance platform that leverages Retrieval-Augmented Generation (RAG) to query relevant subgraphs, reducing hallucinations and improving interpretability. Using the Huawei OpenLineage dataset [Chen et al. \[2024\]](#), we compare DAG-to-RAG against traditional lineage exploration and base LLM, showing its potential to enhance transparency and ease the cognitive burden of analyzing complex data flows.

2. Proposed Solution

This project introduces an **LLM-Guided Data Provenance Platform** designed to capture and explain complex data processing tasks. The platform emphasizes transparency, reliability, and accessibility in data governance.

2.1 Core Components

Benchmark Data: We utilize the open dataset of data lineage graphs (DLG) introduced by [Chen et al. \[2024\]](#).

In this dataset, data assets are represented as nodes (tables, columns, and SQL snippets), while relations are represented as edges (SQL transformations). This structure provides data-level provenance without exposing raw values, making it suitable as a reference benchmark for evaluating lineage capture and explanation.

RAG-to-DAG: An LLM model, enhanced with RAG to incorporate DAG originally suggested by [Hu et al. \[2025\]](#), produces **human-readable summaries** of lineage paths. Relevant lineage segments are retrieved from the stored graph metadata and provided as a context after conversion to natural language. In consequence, each explanation is validated against the DAG structure, to reduce hallucinations and ensure greater faithfulness of the generated summaries.

Lineage Capture: Lineage metadata is ingested and normalized into an **OpenLineage**-compatible schema, and stored in **Marquez** (backend and UI). In addition to the benchmark dataset, we plan to capture OpenLineage-compliant datasets from real-world sources, such as the NYC Taxi dataset [ScienceDirect \[2024\]](#) and the MIMIC-IV dataset [Johnson et al. \[2023\]](#), to demonstrate practical applicability.

This solution enables efficient debugging, supports compliance validation, and enhances transparency for both technical and non-technical stakeholders.

3. Novelty and Contributions

There are three main directions of recent comparable development:

1. Some initial attempts have emerged to embed DAG data in RAG frameworks [Hu et al. \[2025\]](#), but they neither make lineage interpretable for non-technical users nor leverage conversational AI for interactive exploration.
2. Open-source systems such as **OpenLineage**, **Marquez**, and **Spline** capture and visualize lineage metadata, but do not support interactive explanation AI in any form.
3. In machine learning, provenance tools like ML Metadata [Ma et al. \[2024\]](#) and DataHub [Bhardwaj et al. \[2015\]](#) manage artifacts and metadata at scale, yet remain catalogs and trackers without conversational or lineage-aware RAG integration.

Interactive Q&A with Verification: To the best of our knowledge, the proposed *DAG-to-RAG* framework is the first to integrate DAGs into a RAG model while explicitly preserving their structural properties and transformation semantics. This enables stakeholders to interactively explore and understand complex lineage through conversational AI, with reduced risk of hallucination.

Concrete Dataset: In addition, we will be implementing them on real world application with concrete dataset, such as NYC Taxi [ScienceDirect \[2024\]](#) and weather dataset, and the MIMIC-IV dataset [Johnson et al. \[2023\]](#), with which we will perform several ablation studies and present a demo.

Multi-Granular Summaries: Lastly, we are performing Multi-Granular Summaries of several levels: Pipeline → Transform → Column, something not offered by existing tools. Go beyond static lineage visualizations.

In summary, our contributions lie in introducing the first conversational framework that integrates DAG-based lineage into RAG, supported by real datasets and multi-granular summaries, thereby advancing both interpretability and reliability in data provenance.

4. Implementation Plan and Deliverables

4.1 Development Timeline

Stage	Description	Week
Environment Setup	Docker Compose: Spark, Marquez, Postgres, UI	1–2
Lineage Capture	Huawei benchmark dataset <i>OR</i> lineage from ≥ 3 Spark jobs	3–4
Summarization API	Multi-granular, verified summaries	5–6
RAG Q&A	Retrieval + LLM answers with citations	7–10
Evaluation	Overhead $< 15\%$, Hallucination reduction $\geq 30\%$	11–12

4.2 Final Deliverables

Our final deliverables will include a working prototype with a user interface that integrates lineage DAG visualization, multi-granular summaries, and a conversational chat component. We will also provide an evaluation report covering both system overhead and the quality of the generated summaries and Q&A. Finally, we will prepare a slide deck along with a live demo for the final presentation.

5. Expected Impact and Evaluation Metrics

Evaluation Metrics and Targets: System overhead is expected to remain $\leq 15\%$ (runtime comparison). Hallucination rate reduction $\geq 30\%$. Query response time must be ≤ 5 seconds (automated testing).

This project represents a significant step forward in making data lineage transparent, verifiable, and accessible to diverse stakeholders in modern data-driven organizations.

References

- Anant Bhardwaj, Souvik Bhattacharjee, Amit Chavan, Amol Deshpande, Aaron J. Elmore, Samuel Madden, and Aditya Parameswaran. Datahub: Collaborative data science & dataset version management at scale. In *Proceedings of the 7th Biennial Conference on Innovative Data Systems Research (CIDR)*, Asilomar, California, USA, January 2015. URL http://cidrdb.org/cidr2015/Papers/CIDR15_Paper2.pdf. CIDR 2015.
- Yunpeng Chen, Ying Zhao, Xuanjing Li, Jiang Zhang, Jiang Long, and Fangfang Zhou. An open dataset of data lineage graphs for data governance research. *Visual Informatics*, 8(1):1–5, 2024. ISSN 2468-502X. doi: <https://doi.org/10.1016/j.visinf.2024.01.001>. URL <https://www.sciencedirect.com/science/article/pii/S2468502X24000020>.
- Yuntong Hu, Zhihan Lei, Zheng Zhang, Bo Pan, Chen Ling, and Liang Zhao. GRAG: Graph retrieval-augmented generation. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 4145–4157, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics. ISBN 979-8-89176-195-7. doi: 10.18653/v1/2025.findings-naacl.232. URL <https://aclanthology.org/2025.findings-naacl.232/>.
- Alistair E. W. Johnson, Lucas Bulgarelli, Lu Shen, Alvin Gayles, Ayad Shammout, Steven Horng, Tom J. Pollard, Sicheng Hao, Benjamin Moody, Brian Gow, Li-wei H. Lehman, Leo A. Celi, and Roger G. Mark. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, 10(1):1, January 2023. ISSN 2052-4463. doi: 10.1038/s41597-022-01899-x. URL <https://doi.org/10.1038/s41597-022-01899-x>.
- Jiaxuan Ma, Bin Cao, Shuya Dong, Yuan Tian, Menghuan Wang, Jie Xiong, and Sheng Sun. MLMD: a programming-free AI platform to predict and design materials. *npj Computational Materials*, 10(1):59, March 2024. ISSN 2057-3960. doi: 10.1038/s41524-024-01243-4. URL <https://doi.org/10.1038/s41524-024-01243-4>.
- ScienceDirect. Nyc taxi + weather dataset: Integrated mobility and environmental analytics. *Journal of Big Data Applications*, 12(3):145–162, 2024.